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Fixed hybrid dental attachment assembly and methods of use

Abstract

A dental assembly is provided, which includes an endosseous dental implant, a retention member, and a cap configured for securing in a dental appliance and engaging the retention member. The endosseous dental implant may have an external projection extending from the top of the first end portion for releasable engagement with an abutment member, wherein the abutment member has an external lower bevel portion. The retention member may have a first end and a continuous, unbroken skirt extending from the first end defining an inner surface for engagement over the outer locating surface of the abutment member, the retention member being constructed from a rigid material such that it provides a retentive force of about **10** to **75** pounds. The rigid material may include PEEK.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. patent application Ser. No. 17/478,320, filed on Sep. 17, 2021, which is a continuation-in-part of U.S. patent application Ser. No. 17/394,275, filed on Aug. 4, 2021. All references, including publications, patent applications, and patents, cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

TECHNICAL FIELD

(1) Devices and methods provided herein relate to a dental attachment assembly for anchoring a dental appliance with a base structure such as a tooth root or dental implant.

BACKGROUND

(2) Standard dentures are fully removable dentures that are placed in the mouth after all teeth have been removed. Once the mouth tissue has healed, the dentures are placed. These dentures are held

in place by suction contact and/or denture adhesive. For a standard denture to be placed any remaining teeth must be extracted and the patient will be fitted for the device. Without the extracted teeth's previous support, the gum ridge will begin to collapse and the mouth will develop a sunken look. Since the gum ridge flattens over time and the tongue tends to knock standard dentures loose, such a device can become a nuisance at best or even a serious inhibitor to such basic functions as eating and speaking. These and similar minor disturbances in the oral cavity can constitute a major disruption to one's quality of living.

(3) There are generally two types of alternative dentures, fixed or removable dentures, often referred to as overdentures. Overdentures form a gum-supported base onto which a removable denture can be placed; overdenture implants are typically performed on the bottom teeth. Implants are placed into the jawbone to serve as anchors for the overdenture itself. Often the overdenture is made for the bottom teeth, since traditional dentures remain less stable on the lower palate than the upper due to the lower ridge's u-shape, lack of osmotic pressure, tongue movement, and other factors. After the implants are placed, the restoring dentist may use LOCATOR® attachment systems to attach the removable prosthetic denture to the implants. The LOCATOR attachments ensure that the denture will not dislodge unintentionally and will not inconvenience the patient by preventing sliding or shifting in the mouth. Overdentures are designed to be removable by the patient for daily hygiene maintenance.

(4) The advantages of overdentures are manifold. Often people choose implant overdentures over conventional dentures because implants offer a conservative way to add retention, stability, and comfort to the edentulous patient. Instead of sliding freely around the gum area, the denture remains snugly in place in the mouth because it is anchored to the implants. One way this benefits the patient is by allowing her to maintain her dietary health. While someone with traditional dentures must stay away from chewy or hard foods for fear of dislodging the denture, someone with an implant-secured overdenture can maintain his ability to adhere to his regular diet, fully chew his food, and absorb vital nutrition.

(5) Furthermore, the implants themselves help stop facial-contour deterioration, improving not only the physical but also the mental well-being of implant overdenture wearers. Ordinary dentures cause facial bone loss, perpetuating jaw collapse and premature aging. Such facial bone collapse occurs because after one's teeth are removed and one begins to wear a denture, the denture speeds up bone loss by deteriorating the bone ridges where the denture is placed. Moreover, the jaw area increasingly resorbs in the areas where the teeth have been extracted due to lack of stimulation. Overdenture implants provide support for the jaw and gum area, since the implants will naturally integrate with live gum and bone, fortifying the jaw line and preventing bone resorption and facial bone loss. Such osseointegration not only keeps the facial bones physiologically healthy, but it also improves the psychological health of wearers by allowing them to speak, eat, smile, and maintain their natural appearance with increased confidence.

(6) Overall, implant overdentures present a more comfortable, healthy, and durable option than standard dentures. The implants keep the denture secured in place, allowing the patient to converse with confidence, participate in normal social interactions, and even enjoy a diverse diet for optimal nutritional health. Moreover, the implants keep the facial contours and jaw line from collapsing, allowing edentulous patients to maintain bone retention and actually strengthen the jaw line.

(7) But overdentures are not without drawbacks. Dental anchoring or attachment assemblies are utilized to anchor the overdenture appliance with a dental implant or tooth root, typically by fitting two or more partially-movable components together to provide an improved fit and comfort. In some assemblies, male and female parts have mating, snap engageable formations for releasably securing the male part to the female part. For example, the female part, which resides in the appliance or prosthesis, has a socket and the male part, which is anchored to the dental implant or tooth root has a head for snap engagement in the socket. But repeated impacts of the socket and head, as the prosthesis is removed and reattached, can damage the retentive head of the male and

cause wearing due to friction of the components as they move. A compressible annular ring can be provided to absorb the frictional forces and act as a cushion between the socket and the head; however, even the ring may wear out over a period of time and need to be replaced, requiring regular maintenance of the dental anchoring assembly that is uncomfortable and inconvenient for the patient. Furthermore, to allow the compressible annular ring to be easily removed and replaced, the ring can be provided with a securing mechanism on a mating surface with the socket (such as a threaded portion), which further adds to the cost and complexity of the dental anchor assembly.

(8) FIGS. 15A and B illustrate removable overdenture assemblies. As can be seen the prosthesis **1900**, comprises a gingival flange **1910** that covers and is supported by the gums **1908**. Female caps **1902a, b, c, and d** are placed within prosthesis **1900** and mate with male abutments **1904a, b, c, and d** to anchor the prosthesis **1900**. Abutments **1904a, b, c, and d** are typically screwed into implants **1906a, b, c, and d**. The caps **1902a, b, c, and d** form a retentive force with the abutments **1904a, b, c, and d** that holds the prosthesis in place; however, the retentive force is necessarily such that it can be overcome so the prosthesis can be removed for daily cleaning. While four implant assemblies are illustrated in FIG. 15A, it is not uncommon for only two implant assemblies to be used for removable overdentures.

(9) FIG. 15B is another example of a removable denture assembly that uses a bar **1914** to form an implant-supported, removable denture assembly. As can be seen the bar includes, or couples with implants **1906a, b, c, and d**, and can accommodate abutments **1904a, b, c, and d**. Prosthesis **1900**, with flange **1910**, can then “snap” down on abutments **1904a, b, c, and d**. The assembly of FIG. 15A is referred to as an implant-retained denture assembly, while the assembly of FIG. 15B is an implant-supported denture assembly.

(10) As noted above, an annular ring (not shown), or retention head as described below with respect to FIGS. 6-10, can be included within the caps **1902a, b, c, and d**. But issues may still persist as described.

(11) Another problem with removable denture assemblies is that overtime the tissue can recede.

(12) These issues can be addressed with permanent or fixed dentures, which are basically denture devices that consist of a row of prosthetic teeth connected to a framework that is held in position by dental implants and is only removable by a clinician. A typical procedure for placement of permanent dentures involves an oral examination of the patient's dentition, followed by scanning of the jawbone to determine the precise locations for implant placements—two in the anterior region and two at the back of the jawbone at minimum. The permanent dentures will be affixed to the implants following the healing period, during which time the implants would biologically fuse to the jawbone via osseointegration.

(13) FIGS. 16A and B illustrate various example fixed denture assemblies. As can be seen in FIG. 16A, prosthesis **2000** is similar to prosthesis **1900**, but does not include a gingival flange. This is because prosthesis **2000** does not come out and there must be some room to clean under prosthesis **2000** and over gums **2008**. But as with prosthesis **1900**, prosthesis **2000** can be attached to abutments **2004a, b, c, and d**, which can in turn be screwed or otherwise couple with implants **2006a, b, c, and d**. But again, unlike the assembly of FIGS. 15A and B, prosthesis **2000** is typically affixed to the abutments **2004a, b, c, and d** via screws **2012** that screw into abutments **2004a, b, c, and d** form the top of prosthesis **2000**.

(14) FIG. 16B is another example of a fixed denture assembly that uses a bar **2014**, which is common. As can be seen the bar includes, or couples with implants **2006a, b, c, and d**, and can accommodate abutments **2004a, b, c, and d**.

(15) The screws **2012**, abutments **2004a, b, c, and d**, and implants **2006a, b, c, and d**, are metal, which raises the cost, but also result in several metal on metal interfaces that can lead to alignment and seating issues. While the idea is that the fixed appliance **2000** is tightly coupled to the base assemblies, i.e., abutments **2004, b, c, and d**, and implants **2006, b, c, and d**, such that the appliance **2000** cannot move, there are always, e.g., cantilever and other forces that act on the appliance **2000**.

Moreover, it is sometimes necessary to remove a fixed dental appliance, which can create wear on the metal on metal interfaces. Because the metal on metal interfaces are so rigid, and the appliance is not supported by soft tissue as in the case of removable, these forces are dissipated through the weaker appliance **2000**, which can eventually lead to cracking of the appliance **2000**.

(16) FIG. **16B** illustrated another example fixed denture assembly that includes additional metal, screw retained parts **2005**, which can increase costs even further and further exacerbate the problems described.

(17) Another problem with conventional fixed denture assemblies is the need to remove up to 15 mm of vertical bone height to accommodate the stacked components and make room for the prosthesis.

(18) Another problem with conventional fixed denture assemblies is related to the labor intensive clinician appointment for hygiene. Retaining screws must be located, uncovered and removed to free the appliance for cleaning. The process then must be repeated in reverse to reattach the appliance. This procedure has a risk of appliance fracture due to the through holes required for screw access and is often avoided unless another problem requires it. This is a disadvantage for the patient and their overall oral hygiene.

SUMMARY

(19) According to one aspect, a dental attachment assembly comprises an abutment member for attachment to a tooth root, implant, or adjacent tooth, a retention member for fixed, non-patient removable attachment to the abutment member, and a cap configured for securing in the recess in a dental appliance. The abutment member may have an upper end, and an outer locating surface portion projecting downwardly from the upper end, the outer locating surface portion being positioned to project above a tissue level when the abutment member is secured in a tooth root or implant. The abutment member may have a head portion at the upper end and a shaft depending downwardly from the head portion. The shaft may be at a predetermined angle to the head portion (e.g., 10 degrees, 20 degrees, 30 degrees, or aligned with the head portion). The retention member may have an upper end configured for engagement within a cap in a recess in a dental appliance, and a continuous, unbroken skirt projecting from the upper end of the retention member for engagement over the outer locating surface of the abutment member, the skirt having a rounded, convex outer surface, the retention member being constructed from a polyetheretherketone (PEEK) material or other rigid material such that it provides a retentive force of about 10 to 75 pounds. The abutment member and retention member may have mateable snap-engaging formations for releasable snap engagement when the retention member is attached to the abutment member. In some embodiments, the abutment member may have a socket projecting inwardly from the upper end, of the abutment member and a retention head projects from the upper end of the retention member for releasable snap engagement in the socket. The retention head may have a lower end, with the skirt extending downwardly beyond the lower end of the retention head, whereby the skirt will contact the abutment member before the lower end of the retention head reaches the socket on re-insertion of the retention member into the abutment member. The abutment member may have a weep hole connecting the socket to the outside of the abutment member for allowing saliva to escape. In some embodiments, the mateable snap-engaging formations may comprise an outwardly bulging portion of the outer locating surface portion of the abutment member and a corresponding concave ring portion of an inner surface of the skirt of the retention member. The cap may comprise a cavity having a rounded concave inner surface for containing the upper end of the retention member. The concave inner surface of the cavity may allow for releasable snap engagement over the rounded outer surface of the skirt. In some aspects, the cap may comprise a generally cup shaped member having a central longitudinal axis, a top wall and an annular side wall extending away from the top wall, the top wall and side wall having an outer surface shaped to engage with the recess in the dental appliance, and wherein the cavity is configured for swivel engagement with the retention member adapted for non-swivel engagement over the outer locating

surface of the abutment member attached to the tooth root, implant, or adjacent tooth. The top wall may have an interior top surface with a concave recess extending radially in an annular path along an outer annular peripheral portion of the interior top surface, the concave recess forming a curve having a first radius of curvature. The curve of the concave recess may continue along at least part of the inner surface of the side wall towards an open end of the rounded, concave inner surface of the cavity. The curve may be configured to provide swivel engagement with the retention member between a non-swiveled position in axial alignment with the central longitudinal axis of the retention member and a fully-swiveled position in which the central longitudinal axis of the cavity is at an angle to the central longitudinal axis of the retention member. The range of swivel of the cap over the retention member may be between 10-30 degrees, between 15-20 degrees, approximately 30 degrees, or at least 30 degrees.

(20) According to one aspect, a dental attachment assembly, comprising: an abutment member for attachment to a tooth root, implant, or adjacent tooth, the abutment member having an upper end, and an outer locating surface portion projecting downwardly from the upper end, the outer locating surface portion being positioned to project above a tissue level when the abutment member is secured in a tooth root or implant; a male member (also referred to herein as retention member) for attachment to the abutment member, the male member having an upper end comprising a swivel joint for swivel engagement within a cap in a recess in a dental appliance, and a continuous, unbroken skirt projecting from the upper end of the male member for engagement over the outer locating surface of the abutment member, the skirt having a rounded, convex outer surface, the male member being constructed from a rigid material; the abutment member and male member having mateable snap-engaging formations for releasable snap engagement when the male member is attached to the abutment member; the outer locating surface portion being outwardly tapered at least adjacent the upper end of the abutment member for centering the male member over the abutment member as the members are secured together; and a cap for securing in a recess in a dental appliance, the cap having a cavity for containing said swivel joint, the cavity having a rounded, concave inner surface for releasable snap engagement over the rounded outer surface of the skirt, the swivel joint and cap cavity together comprising means for permitting swivelling of the cap over the male member relative to the dental appliance.

(21) According to another aspect, a dental attachment assembly comprises: an abutment member with a first end shaped to attach to a tooth root, implant or adjacent tooth, and a second end with an outer locating surface, wherein the outer locating surface of the abutment member has at least two axially spaced retention portions comprising two generally rounded, annular projections; a retention member having a first end and a continuous, unbroken skirt extending from the first end defining an inner surface for engagement over the outer locating surface of the abutment member, the retention member being constructed from a rigid material (e.g., PEEK) such that it provides a retentive force of about 10 to 75 pounds, wherein the retention member has at least two axially spaced, snap engaging formations on the inner surface of the skirt for releasable snap engagement with the respective retention portions of the abutment member, and wherein the axially spaced, snap engaging formations on the inner surface of the skirt comprise two spaced, annular grooves for snap engagement over the annular projections on the abutment member; and a cap with a rounded, concave inner surface defining a cavity to receive the retention member, wherein the cap is configured for securing in a dental appliance. In some embodiments, the cavity can be configured to receive a swivel engagement with the retention member, for example, wherein an interior top surface of the cavity includes a concave recess extending radially along an outer periphery of the interior top surface of the cavity. In some embodiments, the at least two axially spaced retention portions of the abutment member can comprise two spaced, annular grooves, and the at least two axially spaced, snap engaging formations on the inner surface of the skirt can comprise two generally rounded, annular projections. In some embodiments, the assembly can comprise a central portion of the interior top surface that protrudes vertically into the cavity, and

the retention member can be in direct contact with the central portion. In some embodiments, a curve of the concave recess continues along a side wall of the rounded, concave inner surface defining the cavity. The concave recess can curve at a plurality of varying angles along the side wall and the interior top surface of the cavity. The concave recess can be composed of a series of consecutive tangent circular arcs, which can one or more of increase in size from the interior top surface to the bottom portion of the cap, and comprised of circles of increasing size from a top portion of the cap to a bottom portion of the cap. The concave recess can provide a retention member with an angle correction of up to, for example, approximately 30 degrees. The cap can have any suitable diameter and height, including for example, an outer diameter of approximately 5.4 mm, and a height of approximately 2.3 mm.

(22) According to another aspect, a dental assembly comprises: a) an endosseous dental implant comprising: (i) a first end portion having a top and a bottom, the first end portion comprises: (1) an external projection extending from the top of the first end portion for releasable engagement with an abutment member, wherein the abutment member has an external lower bevel portion; and (2) an annular cuff portion defining an annular seat having a recessed, inwardly tapered bevel surface, wherein the tapered bevel surface surrounds the external projection; and wherein the lower bevel of the abutment member contacts and seats on tapered bevel surface of the annular seat to form a line of connection when the abutment member is releasably engaged with the external projection, wherein the line of connection is located slightly below a patient's gum line when the implant is secured in the mouth of the patient; and (ii) a threaded post which extends from the bottom of the first end portion for direct engagement into the patient's bone or tissue; b) a retention member having a first end and a continuous, unbroken skirt extending from the first end defining an inner surface for engagement over the outer locating surface of the abutment member, the retention member being constructed from a rigid material (e.g., PEEK) such that it provides a retentive force of about 10 to 75 pounds; and c) a cap configured for securing in a dental appliance and engaging the retention member. It is contemplated that the threaded post can be straight, or can be angled relative to at least one of the first end portion and the abutment member. For example, the threaded post can be angled between 10-20 degrees, about 15 degrees, between 25-35 degrees, or about 30 degrees relative to at least one of the first end portion and the abutment member. In some embodiments, the threaded post comprises a single lead thread or multiple lead threads. In some embodiments, the threaded post comprises opposing axial cuts adapted for self-tapping into the patient's bone or tissue. In some embodiments, the dental implant has a maximum outside diameter between about 1.5 mm and about 4.0 mm. In some embodiments, the threaded post comprises two successive threads of different pitch. In some embodiments, the annular cuff portion may be of different heights to accommodate different tissue depths. In some embodiments, the external projection can include a tool-receiving bore extending inwardly from an upper end of the projection.

(23) According to another aspect, a dental assembly comprises: a dental appliance comprising a first recess and a second recess; a first cap for securing in the first recess and a second cap for securing in the second recess; a first abutment member for attaching to a first tooth root, implant, or adjacent tooth, and a second abutment member for attaching to a second tooth root, implant, or adjacent tooth; a first retention member sized for engagement within the first cap, the first retention member having a first end and a continuous, unbroken skirt extending from the first end defining an inner surface for engagement over the outer locating surface of the first abutment member, the retention member being constructed from a rigid material (e.g., PEEK) such that it provides a retentive force of about 10 to 75 pounds; and a second retention member sized for engagement within the second cap. In some embodiments, the second retention member comprise a first end and a continuous, unbroken skirt extending from the first end defining an inner surface for engagement over the outer locating surface of the second abutment member, the retention member being constructed from a rigid material (e.g., PEEK) such that it provides a retentive force of about

10 to 75 pounds. All suitable abutment members, caps, and retention members are contemplated, including any or all of the abutment members, caps, and retention members described above or in the detailed description.

(24) In some aspects of embodiments comprising a dental appliance, the first abutment member comprises an upper end, and the outer locating surface of the first abutment member is positioned to project about a tissue level when the abutment member is secured in the first tooth root, implant, or adjacent tooth, wherein the first retention member comprises an upper end for engagement within the first cap, wherein the first abutment member and first retention member have mateable snap-engaging formations for releasable snap engagement when the first retention member is attached to the first abutment member, the outer locating surface of the first abutment member being outwardly tapered at least adjacent the upper end of the first abutment member for centering the first retention member over the first abutment member as they are secured together, and wherein the first cap comprises a cavity for containing the swivel joint, the cavity having a rounded, concave inner surface for releasable snap engagement over the rounded outer surface of the skirt, the swivel joint and cap cavity together comprising means for permitting swivelling of the first cap over the first retention member relative to the dental appliance. The first abutment member may have a socket projecting inwardly from the upper end of the abutment member and a retention head may project from the upper end of the first retention member for releasable snap engagement in the socket. In some aspects, the outer locating surface may have an outward taper from the upper end of the first abutment member and terminate at a predetermined location above the tissue level when the first abutment member is secured in a patient's mouth. In some aspects, the first abutment member may comprise a generally cylindrical head having a lower end spaced below the upper end of the first abutment member. In some aspects, the skirt of the first retention member may have an inner surface of predetermined height which is concave and does not contact the outer locating surface portion of the first abutment member along at least a major portion of the height of the inner surface when the first retention member and first abutment members are secured together. In some aspects, the first cap can comprise a generally cup shaped member having a central longitudinal axis, a top wall and an annular side wall extending away from the top wall, the top wall and side wall having an outer surface shaped to engage with the first recess in the dental appliance.

(25) In some aspects of embodiments comprising a dental appliance, the first abutment member may comprise a first end shaped to attach to the first tooth root, implant or adjacent tooth, and a second end with the first outer locating surface, wherein the first outer locating surface of the first abutment member has at least two axially spaced retention portions comprising two generally rounded, annular projections, wherein the first retention member has at least two axially spaced, snap engaging formations on the inner surface of the skirt for releasable snap engagement with the respective retention portions of the abutment member, and wherein the axially spaced, snap engaging formations on the inner surface of the skirt comprise two spaced, annular grooves for snap engagement over the annular projections on the first abutment member, and wherein the first cap comprises a rounded, concave inner surface defining a cavity to receive a swivel engagement with the first retention member; wherein an interior top surface of the cavity includes a concave recess extending radially along an outer periphery of the interior top surface of the cavity. A central portion of the interior top surface may protrude vertically into the cavity, and the first retention member may be in direct contact with the central portion. In some aspects, a curve of the concave recess may continue along a side wall of the rounded, concave inner surface defining the cavity. The concave recess may curve at a plurality of varying angles along the side wall and the interior top surface of the cavity. Additionally or alternatively, the concave recess may be composed of a series of consecutive tangent circular arcs.

(26) In some aspects of embodiments comprising a dental appliance, the assembly further comprises an endosseous dental implant. The implant may comprise (i) a first end portion having a

top and a bottom, the first end portion comprises: (1) an external projection extending from the top of the first end portion for releasable engagement with the first abutment member, wherein the first abutment member has an external lower bevel portion; and (2) an annular cuff portion defining an annular seat having a recessed, inwardly tapered bevel surface, wherein the tapered bevel surface surrounds the external projection; and wherein the lower bevel of the first abutment member contacts and seats on tapered bevel surface of the annular seat to form a line of connection when the abutment member is releasably engaged with the external projection, wherein the line of connection is located slightly below a patient's gum line when the implant is secured in the mouth of the patient; and (ii) a threaded post which extends from the bottom of the first end portion for direct engagement into the patient's bone or tissue. The threaded post may be angled (e.g., between 10-20 degrees, about 15 degrees, between 25-35 degrees, about 30 degrees) relative to at least one of the first end portion and the first abutment member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The details of embodiments of the present disclosure, both as to their structure and operation, may be gleaned in part by study of the accompanying drawings, in which like reference numerals refer to like parts, and in which:
- (2) FIG. 1 illustrates one embodiment of a dental anchoring or attachment assembly that can be attached with an implant that can be anchored to a bone or other base structure such as a tooth root, according to an embodiment;
- (3) FIG. 2 illustrates the external surface of the head of a retention member, according to an embodiment;
- (4) FIG. 3 illustrates another view of the external surface of the head of the retention member of FIG. 2;
- (5) FIG. 4 illustrates a dental attachment assembly in an angled, unattached configuration, according to one embodiment;
- (6) FIG. 5 illustrates the dental attachment assembly in the angled orientation shown in FIG. 4 but where the retention member is now snap-fit into the socket of the abutment;
- (7) FIG. 6 illustrates a removable dental attachment assembly for attaching a dental appliance to an implant, according to one embodiment;
- (8) FIG. 7 illustrates a sectional view taken on line 7-7 of FIG. 6, with a retention member shown separated, according to an embodiment;
- (9) FIG. 8 is a view similar to FIG. 7, showing the retention member of FIG. 7 aligning with the abutment member of FIG. 6, according to an embodiment;
- (10) FIG. 9 is a view similar to FIGS. 7 and 8, including a cap and showing the retention member of FIG. 7 coupled to the abutment member of FIG. 6, according to an embodiment;
- (11) FIG. 10 is a sectional view showing a shortened form of the abutment member of FIG. 6, according to an embodiment;
- (12) FIG. 11A illustrates another example embodiment of a removable dental assembly, according to one embodiment;
- (13) FIG. 11B is an isometric view illustration of the cap of the dental assembly of FIG. 11A;
- (14) FIG. 11C is an isometric cutout view illustration of the cap of the dental assembly of FIG. 11B;
- (15) FIG. 11D is a sectional view illustration of the cap and a retention member of the dental assembly of FIG. 11A;
- (16) FIG. 11E is a sectional view illustration of the cap of the dental assembly of FIG. 11A illustrating consecutive tangent circle arcs which are used to create an angle of a concave recess on

an interior wall of the cap;

(17) FIG. 11F is a sectional view illustration of the cap of the dental assembly of FIG. 11A illustrating an angle of the concave recess;

(18) FIG. 11G is a sectional view of the dental attachment assembly of FIG. 11A in a fully swiveled position;

(19) FIG. 12 illustrates a conventional Peek retention member that includes a gap or slot cut into the retention member in order to allow for some resiliency;

(20) FIG. 13 illustrates one embodiment of a dental anchoring or attachment assembly, according to another embodiment;

(21) FIG. 14A illustrates a dental anchoring or attachment assembly with a lower portion configured to be straight;

(22) FIG. 14B illustrates the assembly of FIG. 14A with a lower portion configured to be angled;

(23) FIG. 14C illustrates the assembly of FIG. 14A with another lower portion configured to be angled;

(24) FIG. 15A illustrates a removable overdenture assembly, according to an embodiment;

(25) FIG. 15B illustrates a removable denture assembly that uses a bar, according to an embodiment;

(26) FIG. 16A illustrates a fixed denture assembly, according to an embodiment; and

(27) FIG. 16B illustrates another fixed denture assembly, according to an embodiment;

(28) FIG. 17 illustrates a fixed hybrid dental attachment assembly, according to an embodiment.

DETAILED DESCRIPTION

(29) The detailed description set forth below, in connection with the accompanying drawings, is intended as a description of various embodiments, and is not intended to represent the only embodiments in which the disclosure may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of the embodiments. However, it will be apparent to those skilled in the art that embodiments of the invention can be practiced without these specific details. In some instances, well-known structures and components are shown in simplified form for brevity of description.

(30) The dental attachment assembly described in U.S. Pat. No. 9,931,181 (the '181 patent), which is incorporated herein by reference as if set forth in full, functions similar to a fixed dental attachment system as described above, yet can be more easily removed by a dental professional using a special tool and which overcomes many of the problems of conventional fixed dental attachment assemblies. The dental attachment assembly described in the '181 patent is useful to attach a dental appliance, not intended to be removable by the patient, such as a denture, with an implant and provide a retentive force of about 8 to 75 pounds, while also providing ease of engagement of the retention member with the abutment due to the compressible nature of the materials used and the friction-retained snap-fit of the ball and socket components included within the assembly.

(31) As will be described in detail below, the retention member is formed from a compressible material—such as a polymer or soft metal—to allow the retention member to compress and flex while being attached or detached from an abutment secured to the implant. The compressible and flexible retention member can then be secured with the abutment at a variety of angles, which is often necessary when securing a dental appliance to a plurality of implants extending at different angles across a person's upper or lower mandible. Additionally, the compressible ball eliminates the need for a separate compressible annular ring to be positioned in the socket of the abutment between the retention member and interior abutment walls, as well as the need for a securing mechanism for securing the annular ring to the abutment walls. The dental attachment assembly is therefore easier to manufacture and requires less maintenance once inserted.

(32) FIGS. 1-5 herein are the same as FIGS. 1-5 of the '181 patent. FIG. 1 illustrates one embodiment of a dental anchoring or attachment assembly **100** that can be attached with an implant

(not shown) that can be anchored to a bone or other base structure (not shown) such as a tooth root. The assembly includes an abutment **102** that is secured to the implant and a cap or denture attachment housing (DAH) **104** that is secured in a recess of a dental appliance. A retention member **106** serves to provide the frictionally-retained connection between the cap **104** and the abutment **102**. To this end, the retention member **106** includes a shaft **108** that, e.g., has a threaded end portion **110** in a threaded connection with a corresponding threaded bore **111** in the cap **104**. A second end of the shaft, which interfaces with the abutment **102**, includes a head **112** that is substantially spherical in shape and which is configured to create a frictional fit with an inward projection or barb **120** in a socket **114** found in an upper opening **116** of the abutment **102**.

(33) The head **112** includes a curved surface **118** configured to frictionally engage an annular inward projection **120** in the socket **114**, and a flat lower end face **119** with, e.g., an inwardly directed hexagonal or polygonal shaped recess **121** for engagement with a suitable tool when the retention member **106** is threadably engaged in bore **111** of cap **104**. The inward projection **120** has an undercut **123** that is engaged with the head **112**. The socket **114** does not necessarily need to be curved to match the curved surface **118** of the head **112**. Instead, the head **112** is in contact with the socket **114** only at projection **120** in most or all attachment orientations. In the embodiment of FIG. **1**, the outer surface of the head **112** is convex, while the outer end face **119** of the head is flat in order to provide for a closer fit of the head **112** with the abutment **102**. The inward projection **120** is configured to have a friction fit with the corresponding diameter of the head **112** at the mouth of the socket **114**.

(34) One benefit of the compressible material for the retention member **106** is that the diameter of the head **112** can be altered to increase or decrease the retentive force provided by the frictional-fit or compression of the head **112** engaging with undercut barb **123** of annular projection **120**. The greater the maximum diameter of head **112**, the higher the retentive force, since the inward projection **120** cuts further into the head when fully engaged. A smaller diameter head **112** provides less retentive force. Retention force may also be varied by using different, softer or harder compressible materials for head **112**, as described in more detail in the '210 patent in connection with the modified embodiment of FIGS. 31 to 35C therein.

(35) FIGS. **2** and **3** illustrate the external surface of the head **112** of one embodiment of retention member **106**. The outer ball-shaped or convex curved surface **118** can have a series of flats or flattened portions **119** around the circumference of surface **118** to reduce the amount of friction between the curved surface **118** of the head and the corresponding curved surface or projection **120** of the socket. In an alternative embodiment, no flats are provided and the head has a smooth outer convex surface. A smooth convex surface increases the amount of friction between head **112** and projection **120**, since the projection digs more deeply into the compressible surface of the head in the attached configuration of FIG. **1**. Thus, one, two, or all of the following parameters can be used to vary the retention force of the head in the abutment socket: head diameter, head shape, and the selected compressible material of the head. The retention force may vary from anywhere between about 10 to about 75 pounds, although some embodiments may provide as little as about 1 pound of retention force for use in the initial positioning of the dental appliance and dental anchoring device.

(36) The cap **104** is configured with an annular internal surface **122** which may be curved to engage with a corresponding curved outer surface **124** of the abutment, providing an additional frictional fit for the dental attachment assembly.

(37) In one embodiment, a ball flange **126** is provided on shaft **108** at a predetermined spacing below threaded portion **110**. Flange **126** extends perpendicular to the axial direction of the shaft **108** and acts as a stop by engaging an opposing surface **127** of the cavity in cap **104** when threaded stem **110** is threaded into bore **111**. The ball flange **126** serves to help locate the ball **112** within the socket **114** and cap **104** and prevent vertical movement of the assembly.

(38) In the embodiment described herein, the retention member **106** can be formed from a

compressible or elastomeric material such as a polymer or a soft metal, non-limiting examples of which include polymers such as polyether ether ketone (PEEK), polyoxymethylene or acetal polymers such as Delrin®, and soft metals such as nickel titanium (nitinol), pink TiCN (titanium carbo nitride) or titanium. The soft metal can be a coating on the surface of the head portion in some embodiments. In one embodiment, the surfaces may be coated with a gold nitride coating to reduce friction.

(39) FIG. 4 illustrates the dental attachment assembly in an angled, unattached configuration prior to full insertion of head **118** in socket **116**, **114**, illustrating the varying angles at which the retention member **106** may be snap-fit into the abutment. In practical applications, the implant may protrude from the bone or tooth root at varying angles from the ideal vertical angle due to the structure of the bone or the placement of the implant during surgery. The dental anchoring or attachment assembly therefore corrects any angular displacement by rotation of the head **112** in the socket **114**. In one embodiment, the angle of approach of the head with respect to the abutment may vary up to about 20 degrees in any direction from the vertically-aligned orientation shown in FIG. 1. In combination with another implant also offset at a similar angle, the dental anchoring device may therefore provide as much as about 40 degrees of angle correction.

(40) FIG. 5 illustrates of the dental attachment assembly in the angled orientation shown in FIG. 4 but where the retention member **106** is now snap-fit into the socket **114** of the abutment **102**. As illustrated in FIG. 5, the head **112** may be secured within the socket **114** despite the differential angle. Furthermore, the annular surface **122** of the cap **104** is also still frictionally engaged around the outer curved surface **124** of the abutment with the offset indicated in FIG. 5, and the annular projection **120** is frictionally engaged at an angle around the opposing annular surface portion of head **118**.

(41) While the dental attachment assembly was a huge step forward for fixed dental assemblies in that it eliminated weakening through holes in the appliance and replaced the screwed together interfaces with the retention member **106** made of compressible or elastomeric material such as a polymer or a soft metal, non-limiting examples of which include polymers such as polyether ether ketone (PEEK), polyoxymethylene or acetal polymers such as Delrin®, and soft metals such as nickel titanium (nitinol), pink TiCN (titanium carbo nitride) or titanium, it is not interchangeable with a removable or fixed appliance and still requires expensive component swaps to convert from one to the other, and can have a restoration height of 15-18 mm.

(42) FIGS. 6-10 are recreations of FIGS. 1-5 of U.S. Pat. No. 6,030,219 (the '219 patent), which is incorporated herein by reference in its entirety, and which illustrate a removable dental attachment assembly **10** for attaching a dental appliance to an implant **12**. The design illustrated in FIGS. 6-10 present improvements that overcome the deficiencies with removable assemblies described above. The assembly comprises a female socket member or abutment element **14** and a male retention member or element **16** which has a swivel engagement in a cap **18** secured in a dental appliance or prosthesis. The female element **14** is of relatively strong material such as titanium coated with titanium nitride, pink TiCN (titanium carbo nitride), DLC (diamond like coating) or similar material. Element **14** has an enlarged upper end portion **20** having an upwardly facing socket **26**, and a downwardly threaded shaft or stem **22** for attachment to an implant fixture **12** secured in the bone of an upper or lower dental arch. In FIGS. 6-10, the stem **22** is threaded for engagement in a threaded cavity in an implant fixture **12**. It will be understood that the female element will be provided in multiple thread configurations and diameters for engagement in any of the various implant fixtures currently available on the market.

(43) The upwardly facing socket **26** is of suitable shape and dimensions for releasable snap engagement with a downwardly depending retention head **28** on the male element **16**. The mating shapes of the socket **26** and head **28** are similar to that described in U.S. Pat. No. 5,417,570 (the '570 patent), the contents of which are incorporated herein by reference, and will have a similar action. An outer locating surface **30** of the upper end portion **20** projects upwardly above the tissue

level **32** when the female element is secured to an implant or tooth root, as best illustrated in FIG. 9. Locating surface **30** preferably has a slight inward taper from a position just above the tissue level up to the upper end of end portion **20**. The surface portion **34** below the gum level may be cylindrical or have a slight inward taper such that the diameter at the lower end **35** of upper end portion **14** matches that of the implant **12**, as indicated in FIG. 9.

(44) A pair of diametrically opposed, axially extending notches **36** can be provided on the locating surface **30**. These are designed for engagement with an insertion tool (not illustrated) for holding the implant or female element **14** and threading it into a matching socket in an implant fixture **12**. The female element or abutment **14** also can have a weep hole **38** connecting the socket **26** to the outer surface of element **14**, as best illustrated in FIGS. 6 and 7.

(45) Unlike retention member **106** described above, which must be fairly inelastic in order to provide the retentive force needed for a fixed assembly, the male element **16** is formed of a material having some resilience, preferably nylon plastic that provides for some level of elastic reformation when the denture is removed and reattached.

(46) The upper surface **40** is generally fiat, and a skirt **42** projects downwardly from the upper surface **40** to surround the retention head **28**. The retention head **28** has an outer shape substantially matching that of the socket **26**. Socket **26** has a tapered leading edge or rim **44**, a reduced diameter portion **45**, an enlarged ring groove portion **46**, and a lower chamfer edge portion **48**. Head **28** of the male element has a matching shape with a chamfer **50** at the lower end, an annular ring portion **52** of enlarged diameter for snap engagement in groove portion **46**, a reduced diameter portion **53**, and an outwardly tapered upper end portion **54**.

(47) The skirt **42** has a lower end **56** which projects below the lower end of head **28**, and an inner surface **60** which is tapered, preferably at an angle of around 10°, and is also concave, as best illustrated in FIG. 9. The dimensions of the skirt and the head are such that the lower end **56** of the skirt will contact locating surface **30** before the lower end of the head contacts the female element, as best illustrated in FIG. 8. The outer surface **62** of skirt **42** is convex or rounded, for snap-fit, swiveling engagement in a cavity **64** of corresponding shape in the cap **18**, as best indicated in FIG. 9. This provides a swiveling, rotational movement or hinging action at the cap, where the attachment is secured to the denture or appliance, reducing wear, in a similar manner to that described in the '570 patent. The pivoting of the nylon or resilient male in the metal denture cap **18** allows minor corrections for non-parallel abutments, as well as providing a longer lasting, resilient connection.

(48) The denture or appliance can be removed repeatedly for cleaning, simply by snapping male head **28** out of socket **26**, and can then be re-inserted. On re-insertion in the direction of the arrow **65** in FIG. 8, for example, the user may fail to align head **28** co-axially with socket **26**. In this case, the lower end of skirt **42** will contact one side of the locating surface **30** before the head **28** reaches the socket. Further downward movement in the direction of the arrow **65** urges the male element to one side, in the direction of arrow **66**, to align head **28** properly with the socket. Thus, the locator surface portion **30** of the female element together with the inner locating surface of skirt **42** act to urge the head **28** into proper alignment with the socket prior to snap insertion in the socket. This avoids the problem of the nylon head potentially hitting the tapered rim **44** of the socket if mis-aligned, which would eventually cause damage to the softer head and decrease retention ability. The skirt and locator surface also allow easier location and insertion of the dental prosthesis by the patient.

(49) The female element **14** can be provided in several tissue cuff heights to match the surrounding gingival level **32**. The minimum height of the overall attachment assembly is much lower than that of the arrangement as described in the '570 patent and that of a fixed assembly such as described with respect to FIGS. 1-5. The minimum height for a non-hexed implant abutment will be of the order of 1.75 mm, while the minimum height for a hexed implant abutment will be around 2.75 mm, which will help in cases where the implant is at tissue level. This is lower than the height of

all such attachments currently on the market today, and has the advantage of providing a much lower profile and increased patient comfort. Other heights will also be provided depending on the depth of the tissue. The low profile above the attachment is also better for tooth placement.

(50) By providing external notches **36** for engagement with an insertion tool, rather than an internal hex indent or the like as in the '570 patent, the height of the female element can be reduced. The notches allow the female element to be held firmly for threading into the implant fixture **12**.

(51) The weep hole **38** is located above the tissue level and makes it easier for saliva to escape from socket **26**, reducing hydraulic pressure as a result of the male head engaging in the socket. This also allows easier connection of the male and female elements. Once the head **28** is in snap engagement in the socket, as in FIG. **9**, the skirt will only contact the locator surface portion **30** at the lower end, while the concave inner surface **60** will be spaced from surface portion **30** along the remainder of the attachment, as best illustrated in FIG. **9**. This also enables easier insertion of the head **28** into the socket once the parts are in proper alignment.

(52) FIG. **10** illustrates a modified female element or abutment **70** for attachment to a non-hexed implant (not illustrated). Element **70** will be used with the male element **16** of FIGS. **6-9** in a similar manner to hexed implant abutment **14**, and like reference numerals have been used for like parts as appropriate. In this embodiment, head portion in FIG. **10** is shorter than in the previous embodiment, and the hex indent **72** of FIGS. **6-9** is eliminated, with head portion **20** instead having a fiat, lower annular face **74** which fits against the fiat upper face of the implant.

(53) FIGS. 1-5 of U.S. Pat. No. 10,687,920 (the '920 patent), which is incorporated herein in its entirety as if set forth in full, and recreated here as FIGS. **11A-G** illustrate another example embodiment of a removable dental assembly that possess the advantages of the removable assembly depicted with respect to FIGS. **6-10** and as described in the '210 patent. But the assembly described in the '920 patent also allows for increased swivel adjustment. As shown in FIG. **11A** and more clearly in the isometric views in FIG. **11B** and FIG. **11C**, in one embodiment, an interior top surface of the cap **1108** is configured with a concave recess **1110** extending radially along an outer periphery of the interior top surface of the cavity. The concave recess **1110** allows the cap **1108** to swivel at a much greater angle with respect to the retention member **1106** and abutment **1104**, which makes it easier for a person to secure the dental appliance onto the abutment **1104**. A central portion **1112** of the interior top surface of the cap **1108** protrudes vertically into the cap cavity and minimizes vertical displacement between the retention member **1106** and the cap **1108** that would have occurred without the presence of the central portion **1112**. In this exemplary embodiment, the cap is approximately 1.9 millimeters in height, although one of skill in the art will appreciate that the dimensions of the cap may vary.

(54) In the embodiment illustrated in FIG. **11A**, the retention member **1106** is retained by a skirt (or side wall) that projects from one end of the cap **1108** for releasable engagement of the retention member **1106** with the cap **1108**, as disclosed in U.S. Pat. No. 6,981,871, incorporated herein by reference in its entirety. The outer locating surface of the abutment **1104** has at least two axially spaced retention surfaces for mating, snap engagement with corresponding spaced snap engagement formations on the inner surface of the retention member **1106**, to provide a stacked, external retention between the abutment **1104** and retention member **1106**.

(55) FIG. **11D** illustrates a side cutout view of the cap **1108** and retention member **1106** in a non-swiveled position, which more clearly illustrates how the retention member **1106** fits within the cavity of the cap **1108** when no swiveling has occurred. The retention member **1106** is in contact with the central portion **1112** of the interior top surface of the cap **1108**. Outer side walls of the retention member **1106** are generally curved to match the curved surface of the inner side walls of the cap **1108**, and they may be in direct contact at a portion of the side walls at any one position to allow for ease of swivel without an excessive amount of friction.

(56) In one embodiment, the concave recess **1110** is specifically designed with a curvature **1114** which may be formed using a series of consecutive tangent circular arcs formed from multiple

circles of varying radii, as illustrated in FIG. 11E and FIG. 11F. In the embodiment illustrated herein, circles A, B, and C represent portions of the tangent circular arcs used to derive the curve **1114**, although any plurality of circles may be used. The circles A, B, and C are arranged from large-to-small diameter from a bottom portion **1108B** of the cap to a top portion **1108A** of the cap to form the varying angles of the curve **1114**. As illustrated in FIG. 11G, the curve **1114** is designed to optimize the swivel of the cap **1108** around the retention member **1106** while maintaining engagement of the retention member **1106** within the cap **1108**. Varying diameters of each of the circles may be used as long as the diameters are arranged from large-to-small from a bottom-to-top direction along the interior side wall of the cap. In the exemplary embodiment of FIG. 11E, the concave recess **1110** is formed from circles A, B, and C with diameters of 0.067, 0.083 and 0.186 inches, respectively (i.e. radii of curvature of 0.034, 0.042 and 0.093 inches, respectively). Circle C has a diameter which intersects the concave recess **1110** on an opposing side of the cap, which therefore defines the diameter of the cap cavity. Each curvature of each circle intersects with a different portion of the concave recess **1110** to form a curved surface **1114** of varying degrees of curvature, as shown in FIG. 11F.

(57) FIG. 11G illustrates one embodiment of the retention member **1106** and abutment **1104** in a fully swiveled position within the cap **1108**, where a portion of the retention member **1106** has swiveled into the concave recess **1110** in the cap **1108**. The opposing side of the retention member **106** has swiveled downward to the point that it is still retained within the cap **1108**, which illustrates the complete range of swivel of the cap **1108** in one direction. In this embodiment, the range of swivel S is approximately 30 degrees with a cap having a diameter D of approximately 5.4 millimeters (mm) and a height H of approximately 2.3 mm, although one of skill in the art will appreciate that these dimensions may be varied and still achieve the same degree of rotation. The range of rotation may reasonably extend to approximately 32 degrees (e.g., between 15-20 degrees) and could be extended up to approximately 40 degrees if the dimensions of the cap, retention member and abutment are altered to allow for greater rotation—for example by increasing the height of the cap and reducing the diameter of the abutment.

(58) As with the assembly of FIGS. 6-10, the retention member **1106** is made of a resilient material such as nylon.

(59) It is generally understood that harder, less resilient material such as Peek will not work for the retention member in a removable denture assembly, because whatever material is used, it needs to repeatedly deform and then reform as the dentures are removed and then reinstalled. Peek simply is not resilient enough. Certain manufacturers have used PEEK for the retention member such with the Straumann® Novaloc® retentive system. But as illustrated in FIG. 12, such PEEK retention members must include a gap or slot cut into the retention member in order to allow for increased resiliency.

(60) While using a solid PEEK retaining member in a removable assembly is counter intuitive, and would not work for a removable assembly, testing has shown that the use of a solid PEEK retaining member with caps and abutments as illustrated, e.g., with respect to the embodiments of FIGS. 1-5, 6-10, 11A-G, 13, and 14A-C can actually provide the benefits of such removable assemblies, i.e., lower cost, no surgical plan and the long term consequences thereof, lower restoration height, fewer components, elimination of screws and metal on metal interfaces, and greater alignment margin, in a fixed assembly. In other words, it has been shown that a PEEK retention member with the assemblies described above (e.g., in place of **16**, **106**, **1106**, **1706** and optionally having the same features and shapes of **16**, **106**, **1106**, **1706**) provide the retention needed for fixed applications.

(61) This represents a major leap forward for fixed assemblies as it not only allows for the advantages of removable assemblies, but it provides huge flexibility in restorative plans. In other words, the patient may start with removable and transition to fixed (or vice versa), although a retrofit is likely required as noted below. But maybe more importantly, it allows for a transition in

the event of a failed fixed restoration to a successful removable restoration with minimal procedures and cost. A failed conventional fixed restoration failure can be emotionally and financially devastating as there is currently no ability to transition. The restoration plan can even change on the fly depending on surgical outcomes.

(62) If a user has a removable assembly, then switching to a fixed implementation, then additional implants (e.g., **1904**) may be required. The flange **1910** can be removed in the short term and the same removable prosthesis **1900** can be used at least in the short term, while a fixed prosthesis is made.

(63) As noted above, the placement of the implants is important, as the placement and other factors can result in cantilever forces that can cause the prosthesis to crack. In certain instances, the patient's jaw may not allow an implant to be positioned far enough back in the patient's jaw to sufficiently address the potential for such forces. While the retention member **1106** of the embodiments described with respect to FIGS. **11A-G** can allow for an angular displacement due to the up to 32 degrees swivel engagement (e.g., between 15-20 degrees swivel engagement), this may not be enough. Thus, the implant may need to be angled, such that the abutment is positioned even further back in the jaw.

(64) For example, FIG. **13** illustrates an assembly **1700** in which the implant **1702**, screws into the abutment **1704**, with retention member **1706**, and cap **1708**. Implant **1702** has an annular cuff portion **1722** defining an annular seat, and abutment **1704** has an external lower portion **1742**. But as illustrated in FIGS. **14A-C**, the lower portion **1802** of implant **1702** can be configured to be straight (FIG. **14A**) or at various angles such as 15 degrees (FIG. **14B**), or 30 degrees (FIG. **14C**). Assembly **1700** can be similar to that described in, e.g., FIGS. 27-49 in U.S. Pat. No. 9,314,318, which is incorporated herein by reference as if set forth in full.

(65) FIG. **17** illustrates a fixed hybrid dental attachment assembly **2100**. Assembly **2100** comprises a cap **2108**, retention member **2106**, and abutment member **2104**. Abutment member **2104** is configured for attachment to a tooth root, implant, or adjacent tooth. Retention member **2106** is provided for fixed, non-patient removable attachment to the abutment member. Cap **2108** is configured for securing in the recess in a dental appliance. The abutment member **2104** may have an upper end, and an outer locating surface portion being positioned to project above a tissue level when the abutment member is secured in a tooth root or implant. The abutment member **2104** may have a head portion at the upper end and a shaft depending downwardly from the head portion. The shaft may be at a predetermined angle to the head portion (e.g., 10 degrees, 20 degrees, 30 degrees, or aligned with the head portion). It is contemplated that abutment member **2104** can comprise any suitable abutment member, including any of the abutment members described in connection with other embodiments described herein. For example, in some embodiments the abutment member can comprise a first end shaped to attach to a tooth root, implant or adjacent tooth, and a second end with an outer locating surface, wherein the outer locating surface of the abutment member has at least two axially spaced retention portions comprising two generally rounded, annular projections or two annular grooves.

(66) The retention member **2106** may have an upper end configured for engagement within a cap in a recess in a dental appliance, and a continuous, unbroken skirt projecting from the upper end of the retention member for engagement over the outer locating surface of the abutment member, the skirt having a rounded, convex outer surface, the retention member being constructed from a PEEK material or other rigid material such that it provides a retentive force of about 10 to 75 pounds, for example about 25-75 pounds or about 50-75 pounds. The abutment member **2104** and retention member **2106** may have mateable snap-engaging formations for releasable snap engagement when the retention member is attached to the abutment member. In some embodiments, the abutment member may have a socket projecting inwardly from the upper end, of the abutment member and a retention head projects from the upper end of the retention member for releasable snap engagement in the socket. The retention head may have a lower end, with the skirt extending downwardly

beyond the lower end of the retention head, whereby the skirt will contact the abutment member before the lower end of the retention head reaches the socket on re-insertion of the retention member into the abutment member. The abutment member may have a weep hole connecting the socket to the outside of the abutment member for allowing saliva to escape. As shown in FIG. 17, upper end of the retention member comprises an outer surface and an inner surface, and inner surface may lack a downwardly depending retention head (e.g., 28). The mateable snap-engaging formations may comprise an outwardly bulging portion of the outer locating surface portion of the abutment member and a corresponding concave ring portion of an inner surface of the skirt of the retention member. In some embodiments, the retention member may have at least two axially spaced, snap engaging formations on the inner surface of the skirt for releasable snap engagement with the respective retention portions of the abutment member. The axially spaced, snap engaging formations on the inner surface of the skirt may comprise two spaced, annular grooves for snap engagement over annular projections on the abutment member, or two annular projections for snap engagement with annular grooves on a corresponding abutment member.

(67) The cap **2108** may comprise a cavity having a rounded concave inner surface for containing the upper end of the retention member. The concave inner surface of the cavity may allow for releasable snap engagement over the rounded outer surface of the skirt. In some aspects, the cap may comprise a generally cup shaped member having a central longitudinal axis, a top wall and an annular side wall extending away from the top wall, the top wall and side wall having an outer surface shaped to engage with the recess in the dental appliance. The cavity may be configured for swivel engagement or non-swivel engagement with the retention member, and the retention member may be adapted for swivel engagement or non-swivel engagement over the outer locating surface of the abutment member attached to the tooth root, implant, or adjacent tooth. In some contemplated embodiments, the top wall may have an interior top surface with a concave recess extending radially in an annular path along an outer annular peripheral portion of the interior top surface, the concave recess forming a curve having a first radius of curvature. The curve of the concave recess may continue along at least part of the inner surface of the side wall towards an open end of the rounded, concave inner surface of the cavity. The curve may be configured to provide swivel engagement with the retention member between a non-swiveled position in axial alignment with the central longitudinal axis of the retention member and a fully-swiveled position in which the central longitudinal axis of the cavity is at an angle to the central longitudinal axis of the retention member. The range of swivel of the cap over the retention member may be between 10-30 degrees, between 15-20 degrees, approximately 30 degrees, or at least 30 degrees. The concave recess may curve at a plurality of varying angles along the side wall and the interior top surface of the cavity. The concave recess may be composed of a series of consecutive tangent circular arcs, which can one or more of increase in size from the interior top surface to the bottom portion of the cap, and comprised of circles of increasing size from a top portion of the cap to a bottom portion of the cap. The concave recess can provide a retention member with an angle correction of up to, for example, approximately 30 degrees. The cap can have any suitable diameter and height, including for example, an outer diameter of approximately 5.4 mm, and a height of approximately 2.3 mm.

(68) Assembly **2100** may also include implant **2102**, and similarly to assembly **1700**, the lower portion of implant **2102** can be configured to be straight, or at various angles such as between 15-32 degrees, especially where the implant comprises an external projection extending from the top portion for releasable engagement with an abutment member. For example, implant **2102** can comprise a) an endosseous dental implant comprising: (i) a first end portion having a top and a bottom, the first end portion comprises: (1) an external projection extending from the top of the first end portion for releasable engagement with an abutment member, wherein the abutment member has an external lower bevel portion; and (2) an annular cuff portion defining an annular seat having a recessed, inwardly tapered bevel surface, wherein the tapered bevel surface surrounds

the external projection; and wherein the lower bevel of the abutment member contacts and seats on tapered bevel surface of the annular seat to form a line of connection when the abutment member is releasably engaged with the external projection, wherein the line of connection is located slightly below a patient's gum line when the implant is secured in the mouth of the patient; and (ii) a threaded post which extends from the bottom of the first end portion for direct engagement into the patient's bone or tissue. It is contemplated that the threaded post can be straight, or can be angled relative to at least one of the first end portion and the abutment member. For example, the threaded post can be angled between 10-20 degrees, about 15 degrees, between 25-35 degrees, or about 30 degrees relative to at least one of the first end portion and the abutment member. In some embodiments, the threaded post comprises a single lead thread or multiple lead threads. In some embodiments, the threaded post comprises opposing axial cuts adapted for self-tapping into the patient's bone or tissue. In some embodiments, the dental implant has a maximum outside diameter between about 1.5 mm and about 4.0 mm. In some embodiments, the threaded post comprises two successive threads of different pitch. In some embodiments, the annular cuff portion may be of different heights to accommodate different tissue depths. In some embodiments, the external projection can include a tool-receiving bore extending inwardly from an upper end of the projection.

(69) However, it should be appreciated that assembly **2100** or a component/components thereof (e.g., cap **2108**, retention member **2106**) can be used in connection with any of the components of any of the assemblies and dental appliances described herein.

(70) In some aspects, a dental assembly can comprise a dental appliance and any suitable number of attachment assemblies (e.g., 1, 2, 3, 4, 5). The dental appliance may have at least as many recesses as the number of attachment assemblies included in the dental assembly. Each attachment assembly may comprise a cap, a retention member, and an abutment. Each attachment assembly can optionally comprise an implant. The attachment assemblies of the dental assembly can comprise the same or different attachment assemblies. For example, the attachment assemblies can comprise one or more of any combination of dental assemblies described herein, including the dental assemblies described and shown in FIGS. **1-5**, **6-10**, **11A-G**, **13**, **14A-C**, and **17**, with the retention member of at least one of the attachment assemblies having a first end and a continuous, unbroken skirt extending from the first end defining an inner surface for engagement over the outer locating surface of the first abutment member, the retention member being constructed from a rigid material (e.g., PEEK) such that it provides a retentive force of about 10 to 75 pounds.

(71) It will be understood that the benefits and advantages described above may relate to one embodiment or may relate to several embodiments. Aspects described in connection with one embodiment are intended to be able to be used with the other embodiments. Any explanation in connection with one embodiment applies to similar features of the other embodiments, and elements of multiple embodiments can be combined to form other embodiments. The embodiments are not limited to those that solve any or all of the stated problems or those that have any or all of the stated benefits and advantages.

(72) The preceding detailed description is merely exemplary in nature and is not intended to limit the invention or the application and uses of the invention. The described embodiments are not limited to usage in conjunction with a particular type of dental attachment assembly for anchoring a dental appliance with a base structure such as a tooth root or dental implant. Hence, although the present embodiments are, for convenience of explanation, depicted and described as being implemented in a dental attachment assembly for anchoring a dental appliance with a base structure such as a tooth root or dental implant, it will be appreciated that it can be implemented in various other types of dental attachment assembly for anchoring a dental appliance with a base structure such as a tooth root or dental implant. Furthermore, there is no intention to be bound by any theory presented in any preceding section. It is also understood that the illustrations may include

exaggerated dimensions and graphical representation to better illustrate the referenced items shown, and are not considered limiting unless expressly stated as such.

Claims

1. A dental assembly, comprising: a) an abutment member having an external lower portion; b) an endosseous dental implant comprising: (i) a first end portion having a top and a bottom, the first end portion comprises: (1) an external projection extending from the top of the first end portion for releasable engagement with the abutment member; and (2) an annular cuff portion defining an annular seat; and wherein the external lower portion of the abutment member contacts and seats on the annular seat to form a line of connection when the abutment member is releasably engaged with the external projection, wherein the line of connection is located slightly below a patient's gum line when the implant is secured in the mouth of the patient; and (ii) a threaded post which extends from the bottom of the first end portion for direct engagement into the patient's bone or tissue; c) a first retention member having a first end and a continuous, unbroken skirt extending from the first end defining an inner surface for fixed, non-patient removable, engagement over an outer locating surface of the abutment member, the first retention member being constructed from a rigid material such that it provides a retentive force of between 10 to 75 pounds, such that upon engagement over the outer locating surface of the abutment member, the first retention member is non-patient removable from the abutment member, wherein the rigid material comprises polyetheretherketone (PEEK); d) a cap configured for securing in a dental appliance and engaging the first retention member, and e) wherein the abutment member is sized and dimensioned for engagement with a replacement retention member made of a more resilient material than the first retention member and having a second end and a continuous, unbroken skirt extending from the second end defining a second inner surface for patient removable engagement over the outer locating surface of the abutment member, in place of the first retention member.
2. The dental assembly of claim 1, wherein the threaded post is angled relative to the external projection of the endosseous dental implant.
3. The dental assembly of claim 2, wherein the threaded post is angled between 10-20 degrees relative to the external projection of the endosseous dental implant.
4. The dental assembly of claim 3, wherein the threaded post is angled 15 degrees relative to the external projection of the endosseous dental implant.
5. The dental assembly of claim 2, wherein the threaded post is angled between 25-35 degrees relative to the external projection of the endosseous dental implant.
6. The dental assembly of claim 5, wherein the threaded post is angled 30 degrees relative to the external projection of the endosseous dental implant.
7. The dental assembly of claim 1, wherein the threaded post comprises a single lead thread or multiple lead threads.
8. The dental assembly of claim 1, wherein the threaded post comprises opposing axial cuts adapted for self-tapping into the patient's bone or tissue.
9. The dental assembly of claim 1, wherein the dental implant has a maximum outside diameter between 1.5 mm and 4.0 mm.
10. The dental assembly of claim 1, wherein the threaded post comprises two successive threads of different pitch.
11. The dental assembly of claim 1, wherein the annular cuff portion may be of different heights to accommodate different tissue depths.
12. A dental assembly, comprising: a) an abutment member having an external lower portion; b) an endosseous dental implant having a maximum outside diameter between 1.5 mm and 4.0 mm, and comprising: (i) a first end portion having a top and a bottom, the first end portion comprises: (1) an external projection extending from the top of the first end portion for releasable engagement with

the abutment member; and (2) an annular cuff portion defining an annular seat; and wherein the external lower portion of the abutment member contacts and seats on the annular seat to form a line of connection when the abutment member is releasably engaged with the external projection, wherein the line of connection is located slightly below a patient's gum line when the implant is secured in the mouth of the patient; and (ii) a threaded post which extends from the bottom of the first end portion for direct engagement into the patient's bone or tissue, wherein the threaded post comprises two successive threads of different pitch; c) a first retention member having a first end and a continuous, unbroken skirt extending from the first end defining an inner surface for fixed, non-patient removable, engagement over an outer locating surface of the abutment member, the first retention member being constructed from a rigid material such that it provides a retentive force of between 10 to 75 pounds, such that upon engagement over the outer locating surface of the abutment member, the first retention member is non-patient removable from the abutment member, wherein the rigid material comprises polyetheretherketone (PEEK); d) a cap configured for securing in a dental appliance and engaging the first retention member, and e) wherein the abutment member is sized and dimensioned for engagement with a replacement retention member made of a more resilient material than the first retention member and having a second end and a continuous, unbroken skirt extending from the second end defining a second inner surface for patient removable engagement over the outer locating surface of the abutment member, in place of the first retention member.

13. The dental assembly of claim 12, wherein the threaded post comprises opposing axial cuts adapted for self-tapping into the patient's bone or tissue.

14. The dental assembly of claim 12, wherein the threaded post is angled relative to the external projection of the endosseous dental implant.

15. A dental assembly, comprising: a) an abutment member having an external lower portion; b) an endosseous dental implant having a maximum outside diameter between 1.5 mm and 4.0 mm, and comprising: (i) a first end portion having a top and a bottom, the first end portion comprises: (1) an external projection extending from the top of the first end portion for releasable engagement with the abutment member; and (2) an annular cuff portion defining an annular seat; and wherein the external lower portion of the abutment member contacts and seats on the annular seat to form a line of connection when the abutment member is releasably engaged with the external projection, wherein the line of connection is located slightly below a patient's gum line when the implant is secured in the mouth of the patient; and (ii) a threaded post which extends from the bottom of the first end portion for direct engagement into the patient's bone or tissue, wherein the threaded post comprises opposing axial cuts adapted for self-tapping into the patient's bone or tissue; c) a first retention member having a first end and a continuous, unbroken skirt extending from the first end defining an inner surface for fixed, non-patient removable, engagement over an outer locating surface of the abutment member, the first retention member being constructed from a rigid material such that it provides a retentive force of between 10 to 75 pounds, such that upon engagement over the outer locating surface of the abutment member, the first retention member is non-patient removable from the abutment member, wherein the rigid material comprises polyetheretherketone (PEEK); d) a cap configured for securing in a dental appliance and engaging the first retention member, and e) wherein the abutment member is sized and dimensioned for engagement with a replacement retention member made of a more resilient material than the first retention member and having a second end and a continuous, unbroken skirt extending from the second end defining a second inner surface for patient removable engagement over the outer locating surface of the abutment member, in place of the first retention member.

16. The dental assembly of claim 15, wherein the threaded post comprises two successive threads of different pitch.

17. The dental assembly of claim 15, wherein the threaded post comprises a single lead thread.

18. The dental assembly of claim 15, wherein the threaded post is angled relative to the external projection of the endosseous dental implant.
